

your code in a highly available environment. All the maintenance and administration related work is taken care by Lambda. It also does logging and monitoring of your code.

- **Fault tolerance** – Lambda has built in fault tolerance capability. Since there are no servers to manage, virtually no downtime occurs.
- **Automated scaling** – AWS lambda scales your code automatically. For example, if usage of your code increases, you need not worry about increasing the servers. Lambda will take care of that.
- **Run complex or long tasks easily** – You can coordinate with multiple lambda functions for running a complex or long tasks. AWS Step function will help you to build a workflow and run these tasks.
- **Integrated security model** – you can integrate your Lambda with IAM service provided by AWS. IAM service will take care of access management on behalf of you. Your AWS resources will be secured.
- **Pay per use** – You will be charged only for the time that has been taken for computation by your code.
- **Flexible resource allocation** – You will have the freedom for defining memory allocation for your code. Other resources like CPU, network bandwidth will be allocated accordingly.

When to use Lambda:

Before going with Lambda, you need to understand when you should really use lambda. Below are some use cases appropriate for AWS Lambda.

- If you need to process your data after certain events like shifts in date, some requests made by end-users, data change in your data store. You can go for Lambda.
- Another scenario where Lambda fits perfectly is real time file processing. If some codes need to be executed as soon as a file is uploaded, you should definitely go for Lambda.
- Serverless backend can be built using Lambda when you need to handle a web request or 3rd party API request.
- Lambda can also be used to perform data validation, data sorting and data transformation triggered by some modification in your data store.
- You can make your web application highly scalable and tolerant to faults using Lambda.

Lambda function versioning and aliases:

AWS Lambda comes with an advanced feature of versioning and aliases to